

IndiaSpare.com × WhatsApp Gateway

Frappe Framework • ERPNext • Dual-Language AI Bot • Anti-Ban Engine

1 Overview & Architecture

IndiaSpare.com operates on the **Frappe Framework / ERPNext** tech stack, handling automotive and industrial spare parts catalogs, inventory, sales orders, dispatches, and invoicing. This WhatsApp Gateway package bridges ERPNext with WhatsApp Multi-Device protocol using **Baileys**, providing seamless customer communication with zero Meta API monthly fees.

🛡️ Anti-Ban Human Emulation

Simulates natural human typing delays (35ms/char, randomized 1.5s reading pauses, and WhatsApp *composing* status updates).

🌐 Dual-Language AI Engine

Customer writes in Hindi, Hinglish, Arabic, etc. → Admin sees English. Admin replies in English → Customer receives in their native tongue.

⚡ Frappe Webhooks & Scripts

Instant notification triggers on Sales Order submission, Delivery Note dispatch, Payment Entry receipts, and Lead creation.

💬 Multi-Device LID Resolution

Auto-resolves WhatsApp Multi-Device `@lid` tokens to the customer's actual 10-digit Indian SIM phone number (`+91...`).

Component	Specification / Technology	Purpose in IndiaSpare Stack
Core Engine	Node.js (Express + Baileys v6.7)	Direct WhatsApp Multi-Device Socket connection
Business Phone	<code>+91 73230 00213</code>	Official customer service SIM registered for IndiaSpare
ERPNext Listener	<code>POST /api/webhook/frappe</code>	Receives DocType webhook events from ERPNext
Universal Dispatch	<code>POST /api/send</code>	Allows Frappe Server Scripts (Python) to send messages
Admin Dashboard	<code>http://localhost:3001/admin</code>	Live Inbox, auto-reply rules editor, and QR screen
Web Chat Widget	<code>indiaspare-widget.js / .css</code>	Floating chat injected into <code>indiaspare.com</code>

2 Quick Start & WhatsApp Phone Pairing

1. Clone / Open the Gateway Directory:

```
cd C:\Users\mddaw\.gemini\antigravity\scratch\wa-gateway-indiaspare
```

2. Install Dependencies:

```
npm install
```

3. Start Gateway Server:

```
npm start
```

The server will bind to `http://localhost:3001`.

4. Scan WhatsApp QR Code:

Open browser at `http://localhost:3001/qr`. On phone `+91 73230 00213`, navigate to WhatsApp → **Linked Devices** → **Link a Device** → Scan the QR code.

5. Persistent Authentication:

Session credentials are encrypted and stored in `auth_info_indiaspare/`. The gateway auto-reconnects across restarts without re-scanning.

✓ **Anti-Ban Safety Protocol:** Never send unsolicited bulk messages. The gateway includes automated cooldown timers (15 mins/customer) and natural presence simulation to preserve SIM reputation.

3 Dual-Language AI Auto-Translation

Automotive mechanics and garage managers across India frequently query in **Hindi, Hinglish, Marathi, Gujarati, or Arabic**. This gateway includes a zero-latency bidirectional translation layer:

Inbound Flow: Customer types "मुझे महिन्द्रा बोलेरो का क्लच प्लेट चाहिए" → Stored and displayed to IndiaSpare staff as "I need a clutch plate for Mahindra Bolero" (with original text preserved for reference).

Outbound Flow: Staff types reply in English "Clutch plate is in stock for Rs 3,450. Shall we dispatch?" → Gateway automatically translates and delivers "क्लच प्लेट 3,450 रुपये में उपलब्ध है। क्या हम इसे भेजें?" to the customer's WhatsApp.

Live Translation API

To preview translations in real-time within ERPNext Desk or custom interfaces:

```
POST http://localhost:3001/api/chats/translate
Content-Type: application/json

{
  "text": "swift dzire diesel front brake pads price",
  "targetLang": "hi",
  "sourceLang": "en"
}
```

4 Auto-Reply Keyword Rules & Bot Engine

The gateway automatically intercepts incoming inquiries and replies based on rules configured in

`auto_reply_rules.json` :

Rule	Trigger Keywords	Match Type	Automated Response Action
Greeting	hi, hello, namaste, hey, salam	contains	Welcome to IndiaSpare.com message + prompts to send part model.
Parts & Spares	part, spare, clutch, brake, filter, piston, gear, engine, bearing	contains	Requests Vehicle Model + Part Number + photo for instant quotation.
Order Status	order, status, track, delivery, courier, dispatch	contains	Prompts for Sales Order number and links to tracking portal.
Commercial Pricing	price, rate, cost, quote, discount, gst, wholesale	contains	Explains 100% GST input tax credit policy & tiered bulk rates.
Urgent Dispatch	urgent, emergency, fast, today, breakdown, asap	contains	Flags consignment as High Priority & routes to on-duty dispatcher.
Official Bank	bank, account, upi, payment, razorpay, gpay, neft	contains	Official Zero213 Ventures Pvt Ltd HDFC/ICICI Bank & UPI ID verification.
Working Hours	timing, hours, open, sunday, location, ranchi	contains	Operating hours: Mon-Sat, 9:30 AM – 7:30 PM IST; Ranchi Hub details.
Human Takeover	agent, human, stop, staff	exact	Disables automated bot for 60 minutes & alerts human team.

5 ERPNext & Frappe Framework Integration

ERPNext communicates with the WhatsApp gateway via two methods: **Frappe Webhooks** (UI based) or **Frappe Server Scripts** (Python code).

Method A: Setting Up Frappe Webhooks (No Coding Needed)

1. Log in to your ERPNext instance (indiaspare.com/app) as **Administrator**.
2. In the top Awesomebar, search for **Webhook** → Click **Add Webhook**.
3. Fill the Webhook form with the following settings:

Field Name	Sales Order Setting	Delivery Note Setting
DocType	Sales Order	Delivery Note
Webhook Event	on_submit (or after_insert)	on_submit
Request URL	http://<your-gateway-host>:3001/api/webhook/frappe	http://<your-gateway-host>:3001/api/webhook/frappe
Request Method	POST	POST
Request Structure	Form URL-Encoded or JSON	JSON

Method B: Setting Up Frappe Server Script (Python)

For fine-grained control and dynamic templates, create a Server Script in Frappe:

1. In ERPNext Awesomebar, search for **Server Script** → Click **New**.
2. **Script Type:** DocType Event
3. **Reference DocType:** Sales Order
4. **DocType Event:** Before Submit or After Submit
5. In the Python script editor, paste the following snippet:

```

# Frappe Server Script: WhatsApp Sales Order Confirmation
phone = doc.contact_mobile or doc.mobile_no or getattr(doc, "customer_mobile", None)

if phone:
    items_list = "\n".join([f"• {i.item_name} (Qty: {i.qty})" for i in doc.items])
    message = (
        f"Hello {doc.customer_name}! 🙋\n\n"
        f"Thank you for ordering with *IndiaSpare.com* (Zero213 Ventures)! 🚚⚙️\n\n"
        f"📦 *Order Confirmed:* #{doc.name}\n"
        f"💰 *Total Amount:* ₹{doc.grand_total:,.2f}\n\n"
        f"🔧 *Parts Reserved:* \n{items_list}\n\n"
        f"Our engineering hub is inspecting your parts for dispatch.\n"
        f"_Reply to this WhatsApp chat anytime if you have fitment questions!_"
    )

    try:
        frappe.make_post_request(
            "http://127.0.0.1:3001/api/send",
            data={
                "phone": phone,
                "message": message
            }
        )
    except Exception as e:
        frappe.log_error(f"WhatsApp dispatch failed: {str(e)}", "IndiaSpare WhatsApp")

```

6 IndiaSpare Website Chat Widget Setup

The widget provides a floating chat bubble on `indiaspare.com` that automatically switches between **Customer WhatsApp Launcher** (for guests) and **Live Customer Inbox** (for logged-in ERPNext staff).

Injecting into Frappe Website Settings

1. In ERPNext, navigate to **Website Settings** (search in Awesomebar).
2. Scroll down to **HTML Header & Scripts**:
 - In **Website Script**, paste the contents of `indiaspare-widget.js`.
 - In **Website CSS**, paste the contents of `indiaspare-widget.css`.
3. Set `window.INDIASPARE_WA_URL = "https://your-gateway-url.com";` at the top of the script.
4. Click **Save** and run `bench clear-cache`.

7 Exposing Public URL & VPS Deployment

Option 1: Cloudflare Tunnel (Prototyping & Testing)

```
# Run Cloudflare Tunnel to expose local port 3001
cloudflared.exe tunnel --url http://localhost:3001
```

Copy the generated URL (e.g. `https://your-subdomain.trycloudflare.com`) into Frappe Webhook settings.

Option 2: Linux Production VPS Deployment (Systemd / PM2)

1. Install PM2 Process Manager:

```
npm install -g pm2
```

2. Start and Enable Auto-Start on Boot:

```
pm2 start server.js --name "indiaspare-wa"
pm2 save
pm2 startup
```

3. Nginx Reverse Proxy with SSL (Port 443 → 3001):

```
server {
    server_name wa.indiaspare.com;

    location / {
        proxy_pass http://127.0.0.1:3001;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
    }
}
```

8 Repository & Version Control

The entire source code, configuration files, and widget assets are version-controlled on GitHub:

- **GitHub Repository:** <https://github.com/devprimetek-svg/indiaspares-wa-gateway>
- **Branch:** `master`
- **Authenticated Account:** `devprimetek-svg` (`dev.primetek@gmail.com`)

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